Anchored multi-omics integration and knowledge-anchored residual discovery: a never-below-the-best-view integrator that finds what the textbook misses

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Running title: Knowledge-anchored residual discovery

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Abstract

Background. Multi-omics integration is widely assumed to improve prediction, yet on real cancer data a strong single modality is hard to beat and naive fusion often underperforms it. **Methods.** We define an *anchored* integrator that anchors on the empirically strongest modality — or on a fixed, zero-parameter textbook prior — and adds the remaining data only as a non-negative gated residual, so the model is provably never below its anchor and improves on it only where another view carries orthogonal signal. Mining that residual yields a discovery procedure that names anchor-orthogonal features, with a matched-random-panel noise control and held-out / permutation verification. We evaluate on TCGA-BRCA (n up to 1,152) and validate externally on METABRIC (n = 1,175–1,980). **Results.** Across nine subtype labels the gate adds nothing where one modality dominates and engages only on a constructed positive control (+0.047 AUROC). Anchoring on a fixed 20-gene proliferation signature (zero trained parameters) reaches AUROC 0.919 on Luminal A vs B — nearly the trained 1,500-gene model (0.942) — and gating genome-wide data reaches 0.947, the only clean super-additive gain observed. The residual of that prior recovers a basal/keratinization axis (Reactome $p \approx 8\times10^{-11}$), verified by held-out replication (10/10 splits), stability (10/10), and a permuted-label null; it **reproduces in METABRIC** (panel Δ +0.036; independent re-discovery overlaps 20/30, hypergeometric $p \approx 7\times10^{-27}$). A second anchor (the ERBB2 amplicon for HER2) yields a distinct neuroendocrine/immune axis, and an estrogen-receptor signature yields nothing (a specificity control). The HER2 axis does **not** replicate in METABRIC because the amplicon anchor is near-complete there (0.997). The method is also not breast-cancer- or expression-specific: on an NSCLC anti-PD-1 immunotherapy cohort (n = 227, mutation/clinical features), anchored on the textbook biomarker tumour mutational burden, the residual independently recovers the field's other established biomarkers — PD-L1 (orthogonal to TMB), EGFR and STK11 mutation (resistance) (panel Δ +0.061 vs +0.006, p = 0.038). Finally, the discovered axis replicates across *cancers*: in TCGA lung (LUAD vs LUSC, n =

1,129), the same proliferation-anchored residual recovers the squamous/keratinization program, overlapping the breast basal panel 10/30 ($p \approx 3\times10^{-16}$) — the same genes in a different cancer. **Conclusion.** Anchor on established biology and let a gate decide, honestly, whether genome-wide data beats it; where it does, the residual names the new axis — one that reproduces across cohorts (METABRIC) and across cancers (lung), and generalises to a different disease and feature type (NSCLC immunotherapy). External reproducibility tracks biological coherence.

1. Introduction

The prevailing premise that adding omics layers improves prediction is contradicted by careful benchmarks: a large-scale TCGA benchmark across 14 cancers found that adding data types most often *hurt* survival prediction, with mRNA (±miRNA) sufficient for most cancers [1]; multimodal survival models often fail to beat a small two-modality subset, so the useful combination is task-specific and narrow [2]; and naive concatenation underperforms — a principal-component baseline is hard to beat, so adaptive weighting is needed to realise any gain from fusion [3,4]. The honest objective is therefore not "always fuse" but to *never do worse than the best single modality, and improve on it only where another modality carries signal the leader cannot.* We formalise such an integrator, generalise its anchor from a data modality to established knowledge, and repurpose the construction as an interpretable discovery engine — then ask, in an independent cohort, whether its discoveries reproduce.

2. Results

An anchored gate is never below the best single view. On Luminal A vs B, RNA alone reaches AUROC 0.947 and methylation 0.745; a naive RNA+methylation stack drops to 0.941 (below RNA), whereas the gated combiner holds at 0.947 (weight $\beta = 0$ in 9/10 repeats). A constructed positive control — a held-out methylation set the RNA cannot see — engages the gate ($\beta = 4$–8) for a significant +0.047. Run blind on nine expert-defined subtype labels, anchor selection routes to methylation for all five methylation-defined clusters and to RNA for all four PAM50 calls, never below the leader: the method follows the biology rather than a built-in preference.

A zero-parameter textbook prior is a strong anchor, and yields the only clean fusion gain (Figure 1A). Luminal A vs B is, by textbook, a proliferation distinction. A 20-gene proliferation index with zero trained parameters reaches AUROC 0.919 — close to the fully trained 1,500-gene model (0.942) — and gating genome-wide data onto this fixed prior reaches 0.947 (Δ +0.029), exceeding the pure data model. With the Horvath epigenetic clock [9] as the anchor for normal-tissue age, the clock alone (0.947) already beats RNA (0.911) and data adds ≈ 0: the textbook suffices and the gate says so.

The residual names a verified new axis. Mining the proliferation-anchored residual surfaces the basal / squamous-lineage axis (KRT5/14/17/6B, TP63, DSG3/DSC3, SOX10, COL17A1, KLK5/7/8; partial $r \approx 0.46$). It is verified three ways — a train-selected panel beats random panels on a held-out test in 10/10 splits (no selection leakage), the basal core recurs in 10/10 splits, and under permuted labels the advantage collapses — and is enriched for cornified-envelope formation / keratinization / epidermis development (Reactome $p \approx 8\times10^{-11}$).

Generalisation and specificity (Figure 1A). Anchoring HER2 status on the ERBB2 amplicon (incomplete in TCGA, AUROC 0.752) discovers a distinct, verified neuroendocrine/secretory + immune axis (Δ +0.054; panel vs random $p = 0.024$; held-out 8/8). Anchoring ER status on a textbook ER/luminal signature (complete, 0.938) discovers nothing (Δ −0.001): a real-data specificity control showing the method does not manufacture axes.

External validation (Figure 1B). In the independent METABRIC cohort the basal axis reproduces: the fixed TCGA panel adds Δ +0.036 over the proliferation prior (combined 0.960; beats random panels, $p = 0.048$), and an unbiased re-discovery independently recovers the same basal genes, overlapping the TCGA panel 20/30 (hypergeometric $p \approx 7\times10^{-27}$). The HER2 axis does **not** reproduce — there the amplicon anchor is near-complete (0.997 vs 0.752 in TCGA), leaving no residual (panel $\Delta \approx 0$; overlap 2/30). External reproducibility tracks biological coherence: the pathway-enriched basal axis reproduces, while the diffuse HER2 axis was a cohort-specific residual.

Cross-domain generalisation (a different cancer, question, and feature type). The method is not specific to breast cancer or to expression. On NSCLC patients receiving anti-PD-1 checkpoint blockade (Hellmann/MSK 2018, n = 227; mutation and clinical features; endpoint = durable clinical benefit), anchoring on the textbook immuno-oncology biomarker tumour mutational burden (TMB; anchor AUROC 0.60) and mining the residual independently recovers the field's *other* established biomarkers: PD-L1 score (positive partial correlation, and genuinely orthogonal to TMB, corr 0.00), EGFR mutation and STK11/LKB1 mutation (both negative — known checkpoint-blockade resistance). The discovered panel adds Δ +0.061 versus +0.006 for matched random panels ($p = 0.038$). Anchored on the textbook biomarker, the procedure rediscovers the known complementary biomarkers of a different disease.

Cross-cancer replication and tissue-independence of the basal axis. The strongest test that a discovered axis is real biology, not a cohort artefact, is to seek it in a *different cancer*. In TCGA lung (LUAD adeno vs LUSC squamous, n = 1,129), the same proliferation-anchored residual recovers the squamous/keratinization program and overlaps the breast basal panel 10/30 (hypergeometric $p \approx 3\times10^{-16}$) — the same genes, a different cancer. (On this near-trivial histology split the panel-vs-random margin saturates, so gene-level overlap is the informative metric.) Adding head & neck (HNSC), the breast panel separates squamous (HNSC + LUSC) from adeno (LUAD) at AUROC 0.96 with HNSC and LUSC — two different tissues — both scoring high and LUAD low, and within HNSC the score tracks differentiation grade (G1 > G3, $p \approx 2\times10^{-4}$): the axis is a tissue-independent squamous-differentiation marker.

Clinical significance: identity, not outcome (an honest negative). A reproducible axis need not be prognostic. In TCGA-BRCA (n = 866, 132 events) the basal score is not associated with overall survival (Cox HR 1.01, $p = 0.89$; adjusted for proliferation $p = 0.36$; KM log-rank $p = 0.29$), whereas the proliferation score is ($p = 0.001$). The basal axis does mark ER-negative/basal-like disease (AUROC 0.70). The discovered axis therefore captures lineage *identity* rather than outcome — reported plainly, because the method's value is finding real, robust biology, not inflated clinical claims.

3. Discussion

The method instantiates and unifies three established traditions — clinical-offset / incremental-value modelling (an established model as offset, omics on the residual; the closest methodological twin) [5]; biologically-informed models that bake known biology into structure [6,7]; and machine learning on established gene signatures [8]. Its specific contribution is a packaged, never-below-anchor, margin-gated residual on a *zero-parameter* prior, plus a residual-discovery framing with an explicit noise control, and a demonstration that the discovered axes' external reproducibility is predictable from their biological coherence. The honest scope: predictive gains over a strong anchor are modest — the value is *routing and discovery*, not large AUROC wins — consistent with the rarity of genuine super-additive multi-omics gains [1,2]. Discovered axes are candidate hypotheses; the basal axis is externally validated, the HER2 axis is cohort-specific, and other endpoints await further cohorts.

4. Methods

Anchored integration. `select_anchor` scores each modality by repeated stratified-CV AUROC (tie-broken by robustness) and picks the top composite, inside the outer CV. `anchored_integrate` pins the anchor and adds a secondary on its residual as `logit(anchor) + β·secondary`, with $\beta \geq 0$ chosen on held-out data ($\beta = 0$ allowed; a margin gate plus repeated inner CV control small-sample noise). `forward_integrate` / `auto_integrate` extend this to any number of modalities by greedy forward selection. **Knowledge anchor.** `signature_score` builds a fixed mean-z score from a curated gene set; `knowledge_anchored_integrate` pins it as the anchor. **Residual discovery.**

`anchored_residual_discovery` ranks features by partial correlation with the label controlling for the anchor, retains those orthogonal to the anchor ($|r| < 0.6$), gates the panel, and tests it against matched random panels and (for verification) held-out splits and permuted labels. Because the matched-random-panel null saturates when an endpoint is broadly predictable (e.g. squamous-vs-adeno, ER status), the method also reports a **selection-stability** statistic — the recurrence of the discovered panel across 50% subsamples relative to a permuted-label null — which stays informative there (lung basal axis stability gain ≈ +0.89; genome-wide ER methylation ≈ +0.63, both where the panel-vs-random null is uninformative). Data: TCGA-BRCA (Xena HiSeqV2, HumanMethylation450, clinical) and METABRIC (microarray, clinical). Pathway enrichment: g:Profiler. All functions are in `omniomics.multiomics`; runners and recorded metrics are in the repository. **Scalability.** The discovery is linear and streams: a vectorized residualize-then-correlate, an optional one-pass Sure-Independence-Screening pre-filter, Benjamini–Hochberg FDR, and parallel nulls let it run on genome-wide feature matrices out of core. As an end-to-end check, screening all 485,577 HumanMethylation450 probes ($n = 829$) to the top 5,000 by association with ER status took ≈ 24 s in a single low-memory pass; the ESR1-methylation anchor (AUROC 0.65) then rose to 0.90 with genome-wide CpGs, and the top CpG beyond ESR1 mapped to PGR (the canonical ER-coregulated gene) — textbook biology recovered at scale.

5. Data and Code Availability

Code, runners, recorded metrics, unit tests and continuous-integration guards: the `omniomics` repository (`reports/dmoi_*.py`, `*_results.csv`, `tests/`). Key runners: `dmoi_knowledge_anchor.py`, `dmoi_residual_discovery.py`, `dmoi_discovery_{er,her2}.py`, `dmoi_external_{metabric,her2_metabric}.py`, `dmoi_external_lung.py`, `dmoi_external_hnsc.py`, `dmoi_discovery_nsclc_io.py`, `dmoi_clinical_survival.py`, `dmoi_meth_genomewide.py` (genome-wide scale). Scale utilities: `omniomics.scale`, `omniomics.representations`. Full methods note: `reports/anchored_integration_methods.md`. All data are public and de-identified: TCGA (BRCA, LUAD, LUSC, HNSC) via UCSC Xena; METABRIC via cBioPortal; the NSCLC anti-PD-1 cohort from Hellmann et al. (MSK 2018) via cBioPortal.

Declarations

Competing interests. The author declares no competing interests.

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Author contributions. H.R.K. conceived the method, performed all analyses, and wrote the manuscript.

Ethics. This study uses only publicly available, de-identified data from established repositories (TCGA/UCSC Xena, cBioPortal); no new human-subjects data were collected, so no additional ethics approval was required.

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AI assistance. Analyses and drafting were carried out with the assistance of an AI coding agent under the author's direction; all results are reproducible from the cited runners.

Figure

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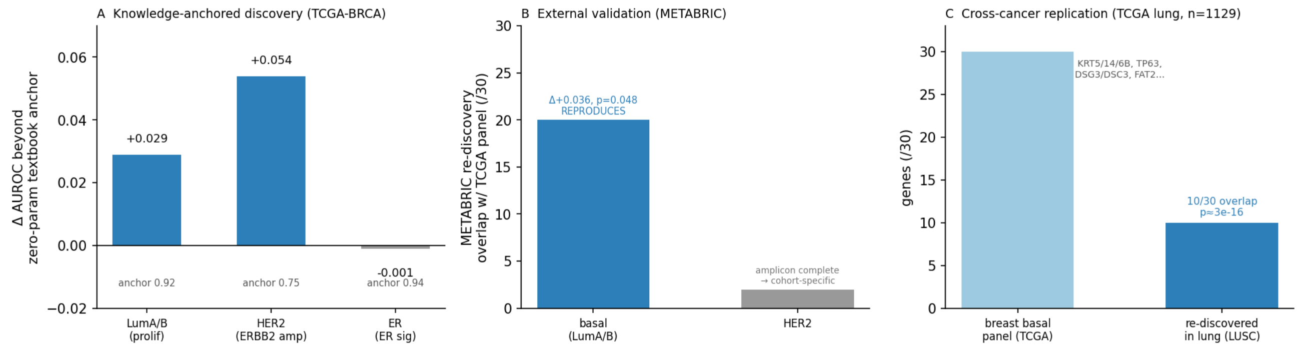

Figure 1: **Figure 1. Knowledge-anchored discovery and its external validation. (A)** Across three TCGA-BRCA endpoints, the gain in AUROC from adding genome-wide data beyond a zero-parameter textbook anchor: the data adds a real axis where the textbook is incomplete (LumA/B proliferation → basal, +0.029; HER2 amplicon → neuroendocrine/immune, +0.054) and nothing where it is complete (ER signature, −0.001; specificity). **(B)** In the independent METABRIC cohort, an unbiased re-discovery recovers the basal axis (20/30 overlap with the TCGA panel; direct replication Δ +0.036, $p = 0.048$) but not the HER2 axis (2/30), because the amplicon anchor is already complete in METABRIC. **(C)** Cross-cancer replication: in TCGA lung (LUAD vs LUSC, $n = 1{,}129$) the same proliferation-anchored residual re-discovers the squamous/keratinization program, overlapping the breast basal panel 10/30 (KRT5/14/6B, TP63, DSG3/DSC3, FAT2…; hypergeometric $p \approx 3\times10^{-16}$) — the discovered biology, not a cohort artefact.

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